

Real-Time Classroom Activity Monitoring Using YOLOv8 and Distributed Streaming Architecture

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Abstract—Classroom Activity Monitoring Systems have emerged as an important application of Artificial Intelligence and Computer Vision in modern educational environments. These systems enable automated analysis of student behavior, such as attentiveness and distraction, through real-time visual data processing. Conventional approaches to classroom monitoring involve manual monitoring techniques. However, such approaches have been found to be subjective and inefficient. The need for data-based information and prompt feedback has created a need for more efficient and accurate monitoring techniques. Various methodologies have been explored to overcome this problem. YOLO-based object detection models have been explored for real-time activity recognition. Convolutional neural networks have been explored for facial expression analysis. Multimodal approaches have been explored for gaze tracking, head pose estimation, and body posture analysis to increase the efficiency of detection techniques. Edge-based inference approaches have been explored to increase the efficiency of the system. Distributed streaming systems such as Apache Kafka have been explored for data processing. Privacy-preserving strategies using structured metadata instead of raw video storage have also been considered. The findings indicate that combining deep learning with distributed architectures improves real-time performance and scalability. Such approaches also ensure efficient monitoring while maintaining privacy in classroom environments.

Keywords—Classroom Activity Monitoring, YOLOv8, Pose Estimation, Distributed Streaming, Real-Time Engagement Analytics

I. INTRODUCTION

The fast-paced rise of Educational Technology has changed face-to-face learning environments from traditional classrooms to digitally-enhanced activities. Despite this technological presence, the majority of ways that educators assess how engaged students are with what is happening in the classroom is still based primarily on teacher observation. Evidence from recent studies has shown that using deep learning-based visual monitoring systems provide a more consistent and reliable approach to measuring student engagement [1].

The majority of the early research examining the automation of classroom processes focused on the area of attendance (i.e., verifying presence). Vision-based systems developed using

facial detection and recognition technology demonstrated that attendance could be accurately recorded without manual roll calling [2]. Additionally, there have been subsequent advances made in the development of these systems such as using face recognition embedded in the Raspberry Pi environment, which increased reliability over single frame detection and allowed for continuous monitoring of students [3]. However, while these systems provided an efficient means of automating attendance taking, they only verified presence and did not evaluate student attentiveness/behavioural engagement.

Behavior recognition systems based on YOLOv5 made it possible to identify attentive and inattentive behaviors in real time across several students, with an accuracy of about 76% in real-world classroom settings [4]. In a similar way, concentration monitoring systems that use deep convolutional neural networks found that they could detect engagement with an accuracy of almost 80% by looking at facial expressions and body language [6]. These advancements collectively validated that object detection frameworks can facilitate real-time classroom monitoring applications.

Following the introduction of fundamental action detection, the next logical step was for researchers to introduce multimodal approaches to detecting user engagement. Through the use of CNN-based facial expression recognition, emotion-aware monitoring models are able to associate affective states (such as concentration and dissatisfaction) with engagement level [8]. Attention tracking systems, in the context of online learning, have used webcam and audio input to compute behavioural metrics relative to engagement and provide adaptive feedback [9]. In addition, a systematic review of published literature relating to the recognition of classroom behaviours further demonstrated that YOLO-based deep learning models have been dominant in more recent implementations, largely due to the potential for real-time inference and ease of deployment [10].

Multi-person posture and motion analysis systems exhibited elevated detection accuracy in densely populated settings; however, scalability continued to pose a challenge [12]. Real-

time engagement scoring systems that use distributed processing methods have made it possible to dynamically index behavior [15]. Lightweight inference frameworks have made edge devices respond more quickly, making it possible for them to respond almost in real time [16]. Transfer learning techniques have further reduced training needs while still providing reliable detection performance in a variety of classroom settings [17]. Large-scale distributed streaming architectures are necessary for AI-driven monitoring solutions to be able to grow and be reliable [18]. Simultaneously, privacy-preserving design strategies—especially those that avoid storing raw video footage and instead keep structured metadata—have been highlighted as essential in educational settings [19].

While the progress made in terms of increasing the accuracy of detection and classifying the level of engagement are impressive, there are still challenges that need attention. Most existing systems operate as singular vision modules and are not connected to larger scalable, distributed streaming architectures. There are also very few systems that have real-time alerting mechanisms to facilitate immediate instruction to students. Issues surrounding data privacy still exist for many systems that keep raw video after having detected behavior. Additionally, limited research exists concerning multi-classroom scalability using fault-tolerant streaming pipeline architectures designed to accommodate large volumes of behavioural events with minimal latency. The limitations of the current systems emphasise the need for an integrated, modular, event-driven framework to provide a comprehensive and deployable solution that incorporates accurate behaviour detection, distributed communication, structured storage, and real-time visualisation.

The proposed Classroom Activity Monitoring System (CAMS) uses a multi-layer event-driven architecture that incorporates Artificial Intelligence, Computer Vision and real-time data streaming technologies to fill in the gaps identified above. The CAMS system uses YOLOv8 to provide pose estimation at the edge layer for a robust real-time activity detection solution. In place of storing raw video streams, CAMS captures and generates structured activity event based on student identifiers, category of detected behaviour, confidence scores, and timestamps. These structured activity events are sent via Apache Kafka to support scalable, distributed, and fault-tolerant communications. FastAPI services provide logic to process incoming streams to the backend, write structured metadata to PostgreSQL using JSONB, and raise real-time alerts using WebSocket communication. A React-based dashboard will display live engagement metrics, distraction trends and classroom-level analytics, allowing educators to monitor classroom dynamics effectively and respond to issues quickly.

The CAMS (Classroom Assistant Monitoring System) is an integrated framework that combines leading-edge object-detection technology with distributed streaming data and privacy-preserving storage functions to create a cohesive solution that unites academic prototype work with real-world operational criteria. While past efforts have primarily focused on improving detection performance, this framework is designed

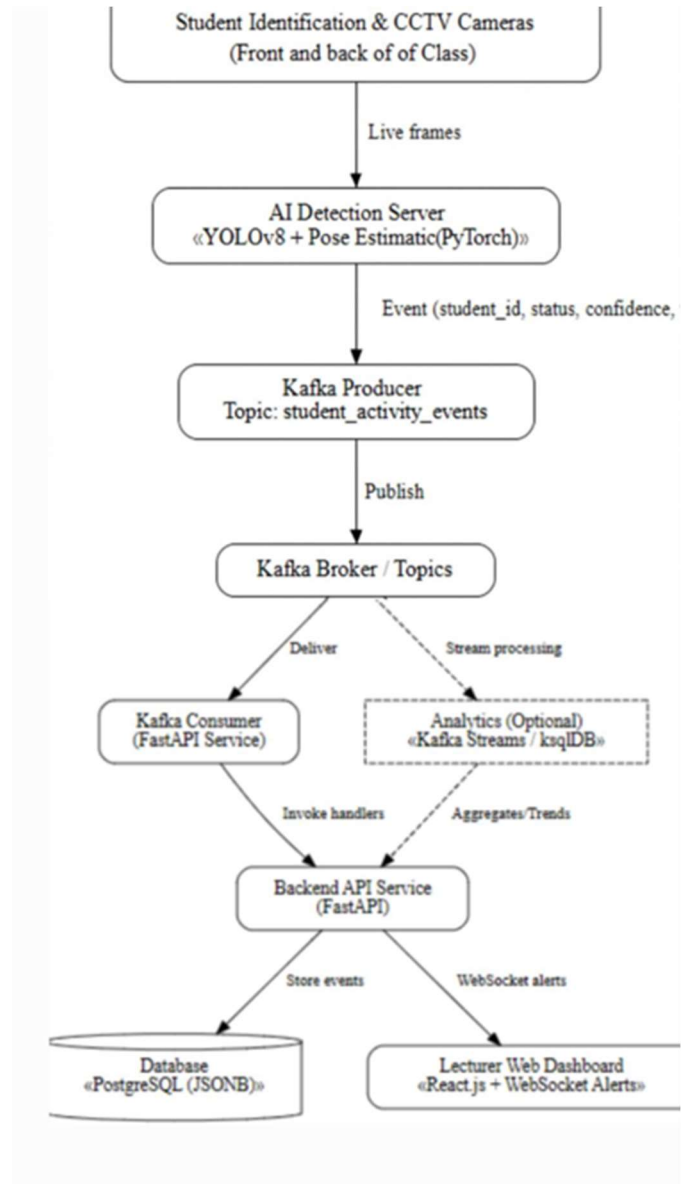


Fig. 1. Overall System Architecture of CAMS

to support deployment in an actual classroom environment by emphasizing scalability, modularity (ability to change), response time, and ethical use of the collected data. Overall, the CAMS framework provides a comprehensive and real-world implementation of intelligent monitoring of classroom engagement, combining state-of-the-art deep learning techniques with the latest principles of modern distributed system design.

II. LITERATURE REVIEW

Over the past decade, the use of Artificial Intelligence and Computer Vision in education has steadily moved from theoretical exploration to practical classroom applications. One of the earliest and most widely adopted research directions was automated attendance management. Systems built using

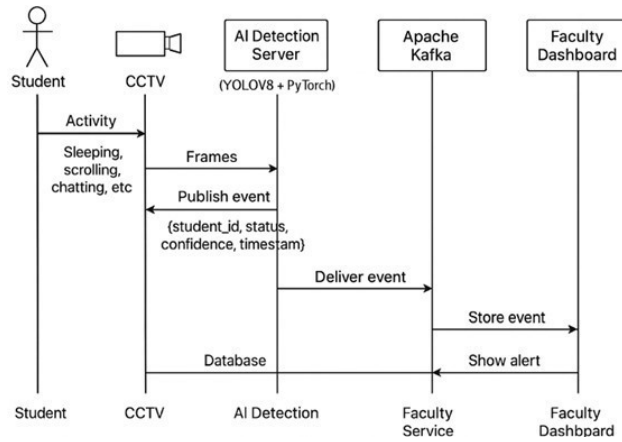


Fig. 2. Detailed Layer-wise Architecture of CAMS

traditional OpenCV-based face detection and recognition techniques, such as the approach described in [9], were designed to remove the need for manual roll calls. However, their scope remained narrow. They verified identity but did not attempt to understand how students were behaving or whether they were engaged in the learning process.

As research progressed, deep learning-based facial biometric systems were introduced to improve robustness under varying lighting conditions, pose variations, and partial occlusions [10]. These advancements enhanced recognition accuracy, especially in real classroom environments. Yet, even with improved reliability, the primary objective of these systems remained attendance tracking. They did not extend toward analyzing distraction, attentiveness, or classroom dynamics.

As deep learning technology has advanced rapidly in the last decade, the research focus has begun to change from "who is present" to "how students behave." For example, Ngoc Anh et al. [1] developed a system that combines face detection with gaze estimation to assess students' levels of attentiveness. In addition to assessing student attentiveness, this work represents a significant step forward in behavioral analytics through producing structured summaries for instructors. However, the system's primary focus was on the accuracy of the detection and did not have any distributed backend or real-time alerting mechanisms.

While Trabelsi et al. [2] created a YOLO-based attention monitoring framework capable of identifying attentive versus non-attentive students and achieving good real-time performance, their model was primarily focused on bounding-box detection and did not address challenges such as scalability, multi-classroom deployment and event-driven system integration. More recent research has shifted to multimodal behavioral analysis to provide a broader view of student engagement. An example of this shift toward using multiple modalities to better understand engagement is in the work by Su et al. [7], who developed a video-based concentration monitoring system that combines facial orientation, pose estimation and activity recognition to achieve approximately 80% accuracy.

Although this system provided a deeper understanding of student concentration and used more data than previous systems, the use of such large amounts of data caused it to require a much larger amount of computing power to operate effectively and did not provide a scalable streaming system architecture that would be able to operate effectively on a large scale in real-time. Likewise, YOLOv5-based multi-student behavior recognition models [14] improved mean Average Precision (mAP) using attention-enhanced feature extraction techniques. However, these approaches largely concentrated on model optimization rather than backend system design or distributed analytics infrastructure.

And as an additional step toward detecting emotion through engagement, researchers have studied using facial expression recognition as a method to measure emotion. In [3], Google researchers proposed a CNN-based model for recognizing facial expressions based on human emotional states such as happiness, dissatisfaction and focus. However, although such systems provide valuable estimates of emotional states, they tend to be very sensitive to the environment in which they are used due to factors such as lighting, occlusions of the face and the angle of the camera. Therefore, different researchers have worked together to develop multimodal frameworks for using eye-tracking, head motion analysis and facial expression recognition technologies to overcome these limitations [18]. However, due to the required controlled environment and their inability to be designed for distributed and real-time deployment across several classrooms, these systems did not help increase the granularity of behavior analysis.

Another avenue that researchers are pursuing to address latency is through the use of edge-based inference models for engagement detection. In this regard Edge-based engagement detection models deployed on edge-based devices provide improved latency performance relative to centralized or cloud-based systems for publishing engagement detection results [12],[17]. However, while these edge-based systems do result in reduced latency with respect to completing engagement detection, they typically did not provide any long-term engagement measures via an integrated analytics platform and were restricted in terms of the behavioral states they were measuring (eg., attentive versus fatigued). At the same time, some researchers developed attention span monitoring systems for online education [19], which were designed to use the web cam and microphone to compute scores of engagement for students in virtual classrooms. Although these systems have demonstrated to be effective in virtual classrooms for distance learning, they are not optimally designed for in-person classrooms where multiple students interact simultaneously.

Research in areas other than education shows that surveillance-based anomaly detection systems such as [8] have advanced recognition of human activity. The focus of these systems is in the area of security rather than the analysis of engagement in learning activities. Liu et al. conducted a systematic literature review [4] about classroom behavior recognition systems and finds that YOLO based and CNN based systems are the most prevalent in the literature. They

also found that a large number of issues were encountered while developing classroom behavior recognition systems including a lack of publicly available data sets, inconsistent labelling procedures, privacy concerns, and no standardized framework for deploying at scale.

Most existing studies focus on improving detection accuracy, while limited attention has been given to scalable architectures, real-time processing, and privacy-preserving system design. This highlights the need for integrated and deployable classroom monitoring frameworks. Much of the existing classroom behavior recognition systems are designed as either independent detection modules that are not integrated with other systems or as centralized systems that do not utilize streaming platforms, fault tolerant strategies, or structured metadata driven storage systems. This imbalance reflects the need to expand beyond model-driven innovation to develop a complete system level design for intelligent classroom monitoring systems.

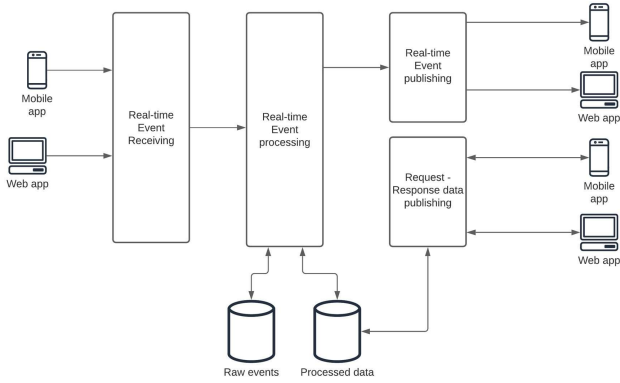


Fig. 3. End-to-End Data Flow Architecture

CAMS is designed to take advantage of the previous research by integrating a YOLOv8-based pose-enhanced behavior recognition approach with a distributed event-driven architecture. Whereas earlier systems tended to record large amounts of raw video and process these video streams in isolation, CAMS provides a structure to the raw activities collected and transmits them for scalable streaming and backend processing using Apache Kafka. The consumption of those events by FastAPI, the storage of the events in PostgreSQL using the JSONB format, and the sending of event notifications to the person executing the activity through a WebSocket connection provide a means for immediate lecturer intervention while still protecting student privacy by not storing the raw video data collected. Furthermore, integrating deep learning inference, distributed streaming, scalable backend infrastructure, and real-time visualization into one system eliminates many of the limitations identified in the existing literature and moves classroom monitoring towards an intelligent method.

Despite these advancements, most existing systems focus primarily on improving detection accuracy while neglecting scalable system design, real-time processing, and privacy preservation. This highlights a clear research gap in developing

an integrated, distributed, and deployable classroom monitoring framework.

III. RESULTS

For assessing Classroom Activity Monitoring System (CAMS), a video was used, which was 50 minutes long and had a size of 337 MB. The video was used for assessing the detection accuracy, on-line performance, and scalability in real-world conditions. For this purpose, Docker Compose was used for building a containerized architecture using Zookeeper, Kafka, PostgreSQL (version 15), and FastAPI for building the backend architecture for event streaming and event storage. All the training and inference were performed using a Tesla T4 GPU with 15 GB VRAM.

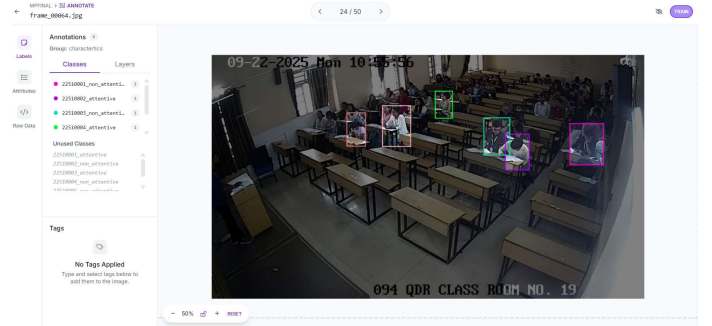


Fig. 4. Classroom CCTV capturing student activities for real-time monitoring.

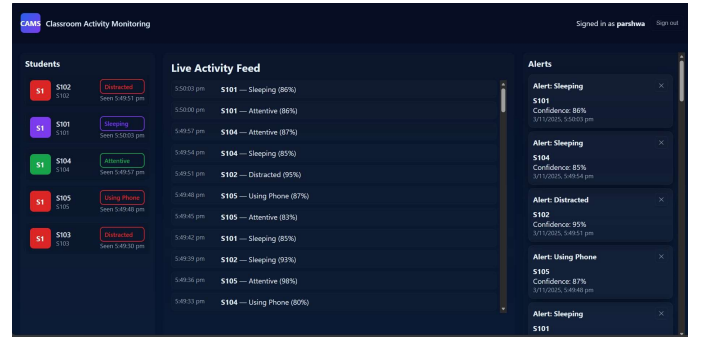


Fig. 5. Real-time CAMS dashboard showing live activity feed and alerts.

A. Detection Performance

The YOLOv8n model, being a lightweight model, was trained for 200 epochs and was able to classify student activities in a classroom setting. The model classified 12 classes based on six student IDs ranging from 22510001 to 22510006, categorized as attentive and non-attentive states. The macro average precision and recall of the model were 83.1% and 83.5%, respectively, with an mAP@50 of 86.1%, proving the effectiveness of the model in the classification of student activities in real-time classroom settings. The model had an mAP@50–95 of 36.2%, proving the reduced localization accuracy due to complex classroom settings involving occlusions, dense seating, and diverse body postures. However, this limitation did not significantly impact

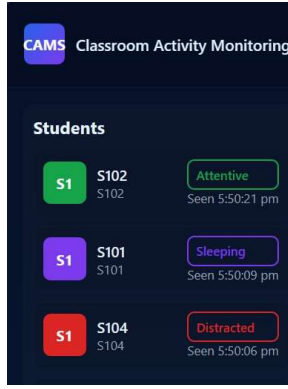


Fig. 6. Student-wise activity status visualization.

the model's ability to accurately classify student behavior, which was the primary objective of the system.

B. Real-Time System Performance

Using a 640x640 resolution during testing, the system provided significant real-time performance. The overall latency per frame from the time processing started to the time processing ended was only 3.8 milliseconds. This indicates that the system could achieve up to 263 frames per second throughput, and as such, it far exceeds the requirements for real-time processing speed for classroom monitoring where timely feedback is essential. Event consumption was handled via FastAPI. Data streaming was accomplished through the use of Kafka with an asynchronous producer (aiokafka) to provide fast consumption of event and status updates based on activity events that were detected. As a result, alerts were sent through WebSockets with an end-to-end latency of less than 100 milliseconds due in part to the system's low, consistent latency, which allowed the receipt of events to occur with an average latency of approximately 7.8 milliseconds. Consequently, the system can be used to monitor classrooms in real-time by providing detailed feedback quickly.

C. Scalability and Storage Efficiency

The system's proposed method has one primary benefit: storage optimization (337 MB of raw video data vs. less than 10 MB for structured activity logs). This represents an estimated savings of more than 97% due to the type of storage used (JSONB). Because of the way the system processes video and activity data, many more classrooms can implement the solution without additional infrastructure expense. Additionally, we do not store video, or any facial images, which aids the overall goal of creating a learning environment in which individual privacy is maintained.

D. Summary

Overall, the CAMS prototype shows good performance in terms of detection accuracy and real-time processing capability. The $mAP@50$ of 86.1% with an inference time of 1.8 ms shows that the system has good accuracy as well as efficiency in processing the data in real-time. The combination of YOLOv8n with the distributed architecture and containerized

TABLE I
PERSON-SPECIFIC PERFORMANCE BREAKDOWN ($mAP@50$)

Student ID	Attentive State ($mAP@50$)	Non-Attentive State ($mAP@50$)
22510001	73.2%	87.6%
22510002	95.3%	92.6%
22510003	77.4%	81.8%
22510004	73.8%	90.3%
22510005	85.5%	96.1%
22510006	87.5%	91.6%

solution makes the system efficient in real-time monitoring, showing the effectiveness of the CAMS solution as applicable in real-world scenarios for classroom engagement analysis.

IV. CONCLUSION

The proposed framework demonstrates that effective classroom activity monitoring extends beyond standalone detection models and requires a tightly integrated system combining deep learning with distributed streaming technologies. By leveraging YOLOv8 with an event-driven architecture, the system enables scalable, low-latency, and privacy-aware analysis of student engagement in real-world environments. The integration with Apache Kafka, FastAPI, PostgreSQL with JSONB, and React WebSocket provides efficient processing, communication, and visualization within a single pipeline. The experimental evaluation demonstrated that the proposed approach achieved $mAP@50 = 86.1\%$, $mAP@50-95 = 36.2\%$, with an end-to-end latency of less than 100 ms. The system was shown to be highly scalable and offered storage reduction by more than 97% through structured metadata storage, which removed the need for raw video data. This demonstrates that classroom monitoring should be viewed as a problem not just from the perspective of computer vision, but as a real-time distributed systems problem, which can be used to create scalable, deployable, and privacy-preserving solutions for intelligent educational environments. Future directions will be to improve the system's robustness in complex classroom scenarios, multi-classroom support, and adaptive feedback mechanisms for better learning outcomes.

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REFERENCES

- [1] 2017 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference. (2017). IEEE.
- [2] M. -C. Su, C. -T. Cheng, M. -C. Chang and Y. -Z. Hsieh, "A Video Analytic In-Class Student Concentration Monitoring System," in IEEE Transactions on Consumer Electronics, vol. 67, no. 4, pp. 294-304, Nov. 2021, doi: 10.1109/TCE.2021.3126877.
- [3] Abdulkader, R., Tayseer Mohammad Ayasrah, F., Nallagattla, V. R. G., Kant Hiran, K., Dadheech, P., Balasubramaniam, V., Sengan, S. (2023). Optimizing student engagement in edge-based online learning with advanced analytics. Array, 19. <https://doi.org/10.1016/j.array.2023.100301>
- [4] Anh, B. N., Son, N. T., Lam, P. T., Chi, L. P., Tuan, N. H., Dat, N. C., Trung, N. H., Aftab, M. U., Dinh, T. van. (2019). A Computer-vision based application for student behavior monitoring in classroom. Applied Sciences (Switzerland), 9(22). <https://doi.org/10.3390/app9224729>
- [5] AUTOMATED CLASSROOM MONITORING WITH IOT AND VIRTUAL APP. (n.d.). www.garph.co.uk
- [6] Fakhra, S., Baber, J., Bazai, S. U., Marjan, S., Jasinski, M., Jasinska, E., Chaudhry, M. U., Leonowicz, Z., Hussain, S. (2022). Smart Classroom Monitoring Using Novel Real-Time Facial Expression Recognition System. Applied Sciences (Switzerland), 12(23). <https://doi.org/10.3390/app122312134>

- [7] Gomes, C., Chanchal, S., Desai, T., Jadhav, D. (2020). Class Attendance Management System using Facial Recognition. ITM Web of Conferences, 32, 02001. <https://doi.org/10.1051/itmconf/20203202001>
- [8] Kamlesh Kahar, Sanjay Satal. (2025). Smart Classroom Attendance System Using Facial Biometrics. International Journal of Scientific Research in Science, Engineering and Technology, 12(5), 01–07. <https://doi.org/10.32628/ijrsrset2512545>
- [9] Katara, A., Sudesh, M., Kolhe, V., Amar, M., Zilpe, P., Nikhil, M., Bhele, D., Chetan, M., Bele, J. (2017). International Journal on Recent and Innovation Trends in Computing and Communication Attendance System Using Face Recognition and Class Monitoring System. <http://www.ijritcc.org>
- [10] Lao, W., Han, J., de With, P. H. N. (2009). W. Lao et al.: Automatic Video-Based Human Motion Analyzer for Consumer Surveillance System Automatic Video-Based Human Motion Analyzer for Consumer Surveillance System.
- [11] Liu, Q., Jiang, X., Jiang, R. (2025). Classroom Behavior Recognition Using Computer Vision: A Systematic Review. In Sensors (Vol. 25, Issue 2). Multidisciplinary Digital Publishing Institute (MDPI). <https://doi.org/10.3390/s25020373>
- [12] Ojo, O. E., Kareem, M. K., Samuel, O., Ugwunna, C. O. (2022). An Internet-of-Things based Real-time Monitoring System for Smart Classroom. Journal of the Nigerian Society of Physical Sciences, 4(2), 297–309. <https://doi.org/10.46481/jnsps.2022.573>
- [13] Quantum Anomaly Recognition. (n.d.).
- [14] Shaheen, J. A. (2017). Apache Kafka: Real Time Implementation with Kafka Architecture Review. International Journal of Advanced Science and Technology, 109, 35–42. <https://doi.org/10.14257/ijast.2017.109.04>
- [15] Sharma, P., Joshi, S., Gautam, S., Maharjan, S., Khanal, S. R., Reis, M. C., Barroso, J., Manuel De, V., Filipe, J. (n.d.). Student Engagement Detection Using Emotion Analysis, Eye Tracking and Head Movement with Machine Learning.
- [16] Subramanian, M., Shanmuga Vadivel, K., Hatamleh, W. A., Alnuaim, A. A., Abdelhady, M., Sathishkumar, V. E. (2022). The role of contemporary digital tools and technologies in COVID-19 crisis: An exploratory analysis. Expert Systems, 39(6). <https://doi.org/10.1111/exsy.12834>
- [17] Trabelsi, Z., Alnajjar, F., Parambil, M. M. A., Gochoo, M., Ali, L. (2023). Real-Time Attention Monitoring System for Classroom: A Deep Learning Approach for Student's Behavior Recognition. Big Data and Cognitive Computing, 7(1). <https://doi.org/10.3390/bdcc7010048>
- [18] Wang, Z., Yao, J., Zeng, C., Wu, W., Xu, H., Yang, Y. (2023). Learning Behavior Recognition in [19] Smart Classroom with Multiple Students Based on YOLOv5. <http://arxiv.org/abs/2303.10916>
- [19] Prateek Rasalkar, Javed Patel. (n.d.). Face Recognition Based Smart Attendance System. <https://doi.org/https://doi.org/10.21203/rs.3.rs-4396816/v1>